WEAKLY ETALE ALGEBRAS OVER A NORMAL DOMAIN ARE INTEGRALLY CLOSED IN THE BASE CHANGE TO THE FRACTION FIELD

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Editorial note. The counted statement span contains the author's two-line cartesian diagram, delivered verbatim. The six references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- A presentation of the fraction field of a normal domain by a product of valuation rings and their fraction fields. invoked at proof line 30735. Supplies the cartesian diagram that makes the integral-closedness question computable.
- Permanence of weak etaleness and of flat injective epimorphisms under flat base change. used at proof lines 30743-30749. Gives the two structural properties of the base-changed rings.
- A ring of weak dimension at most one is integrally closed in the target of a flat injective epimorphism. invoked at proof lines 30750-30751. Is the criterion providing integral closedness on the base-changed diagram.

Source reference. Original lemma 04VQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6333–6333. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00HI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 644–644. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 30303: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30303–30303. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 30679: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30679–30679. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 30706: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30706–30706. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 30231: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30231–30231. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let A be a normal domain with fraction field K. Let $A \to B$ be weakly étale. Then B is integrally closed in $B \otimes_A K$.*

Proof. Choose a diagram as in Lemma 30679. As $A \to B$ is flat, the base change gives a cartesian diagram

$$\begin{array}{ccc} B & \longrightarrow & B \otimes_A K \\ \downarrow & & \downarrow \\ B \otimes_A V & \longrightarrow & B \otimes_A L \end{array}$$

of rings. Note that $V \to B \otimes_A V$ is weakly étale (Lemma 30303), hence $B \otimes_A V$ has weak dimension at most 1 by Lemma 30231. Note that $B \otimes_A V \to B \otimes_A L$ is a flat, injective, epimorphism of rings as a flat base change of such (Algebra, Lemmas 00HI and 04VQ). By Lemma 30706 we see that $B \otimes_A V$ is integrally closed in $B \otimes_A L$. It follows from the cartesian property of the diagram that B is integrally closed in $B \otimes_A K$. □